REACTIVE POWER COMPENSATION SIZING TOOL

A standalone Excel version of the reactive power sizing app. Sheets:
1. Sizing — the calculation chain: grid demand, transformer consumption, losses, inverter sizing,
2. GridCodes — database of published grid-code reactive requirements; pick one from the dropd
3. InverterPQ — the manufacturer's Q(t)-mode capability table (%% of rated kVA per voltage row
4. Compliance — V-Q at Registered Capacity and P-Q at nominal voltage vs the selected grid cod

Colour key: yellow = input cells, grey = calculated, green = key results.

Notes:
- Inverter count: leave the manual-count cell blank to auto-size (meets both reactive and delivery
- Reactive capability precedence: capability table (if 'Use table' = Yes) > manual kVAr override >
- Compliance dispatches inverters at (grid limit + losses) / count, as in a G99 Type C study.
- Export capability is net of transformer consumption; absorption is aided by it; capacitor bank ex
- Grid-code presets are typical published values - always confirm against your interconnection ag
- This tool is an analytic screening aid, not a substitute for a network load-flow study.

PROJECT	
Project name	Belvior

STEP 1 — INITIAL SIZING	
Grid injection limit / Registered Capacity	52.8
Power factor at grid	0.95
MVA rating of plant	55.58
MVAr demand (grid PF)	17.35

STEP 2 — TRANSFORMER REACTIVE CONSUMPTIC	
Group	**Qty**
TS Block 1	3
TS Block 2	4
Auxiliary transformer (MCR)	1
Design margin on transformer demand	10
Transformer demand incl. margin	6.25
TOTAL REACTIVE DEMAND	**23.60**

STEP 3 — INTERNAL SYSTEM LOSSES	
Loss source	**Fraction (%)**
AC cable loss	1
Auxiliary system loss (string inverters)	1
Transmission line loss	0.3
Miscellaneous losses	1
Total system losses	1.742
Capacity to be considered	54.54

STEP 4 — INVERTER CAPABILITY & COMPENSATIC	
Inverter apparent power	363
PF at inverter level (use Data > What-If > Goal Seek on B4	0.918
Active power / inverter	333.2
Reactive capability - derived sqrt(S^2-P^2)	144.0
Reactive capability - manual override (optional)	
Reactive capability - from InverterPQ table (at PF loading)	129.8
Q per inverter USED	129.8
Capacitor bank	0
Q for inverters (after cap bank)	23.60
Manual inverter count (blank = auto)	
Required inverters - reactive	182
Required inverters - active delivery	164
Inverters in design	182
Power delivered to grid (net of losses)	58.91
Reactive power compensated	23.62

Delivery check (>= grid limit)	PASS
Reactive check (>= total demand)	PASS

MW
cos phi
MVA
MVAr

Rating (kVA	Impedance (%)	Q (kVAr)
10164	9.5	2896.7
7260	9.5	2758.8
500	4.5	22.5
		0.0
		0.0
		0.0

%
MVAr
MVAr

Loss (MW)
0.528
0.528
0.158
0.528

MW
MW

kVA
cos phi
kW
kVAr
kVAr
kVAr
kVAr
MVAr
MVAr
nos.
nos.
nos.
nos.
MW
MVAr

Selected gr United Kingdom - ENA G99 / GB Grid Code (Type C/D PPM)

Key	Name
gb_g99	United Kingdom - ENA G99 / GB Grid Code (Type C/D PPM)
de_4110	Germany - VDE-AR-N 4110 (medium voltage)
de_4120	Germany - VDE-AR-N 4120 (high voltage)
us_1547b	USA - IEEE 1547-2018 (Category B)
au_ner	Australia - NER S5.2.5.1 (Automatic access)
eu_rfg	EU - ENTSO-E RfG (typical Type C/D implementation)
pf95	Generic - fixed PF 0.95 lead/lag
pf90	Generic - fixed PF 0.90 lead/lag

Active parameters (looked up from the selected code):

	Q/Pmax
	0.329

Q/Pmax	PF	pMin %	pFullLag %	Relief	Stub	VFullLo	VFullHi
0.329	0.95	20	20	1	0.05	0.95	1.05
0.329	0.95	20	20	0	0	0.95	1.05
0.411	0.925	20	20	0	0	0.95	1.05
0.44	0.9	20	20	0	0	0.95	1.05
0.395	0.93	0	0	0	0	0.95	1.05
0.329	0.95	20	50	0	0	0.95	1.05
0.329	0.95	20	20	0	0	0.95	1.05
0.484	0.9	20	20	0	0	0.95	1.05

PF	pMin %	pFullLag	Relief	Stub	VFullLo	VFullHi	Asym
0.95	20	20	1	0.05	0.95	1.05	1

Asym	VZeroLo	VZeroHi
1	0.9	1.1
0	0.925	1.075
0	0.9	1.1
0	0.9	1.1
0	0.9	1.1
0	0.9	1.1
0	0.9	1.1
0	0.9	1.1

VZeroI	VZeroHi
0.9	1.1

Notes
0.95 lead/lag at RC from 20% output (G99 13.5.5, Fig 13.13/13.14). Asymmetric V-Q; lead relief tr at DNO discretion; +/-5% band below 20%.
cos phi 0.95 under/overexcited above 20% of installed capacity; control range 0.925-1.075 pu.
cos phi 0.925 variant (three TSO-selectable variants exist).
Inject/absorb 44% of nameplate kVA (~0.90 PF at rated); full capability above 20% of rated power
+/-0.395 x rated active power at any output level.
Regulation 2016/631 leaves envelopes to each TSO - typical national choice shown.
Fixed PF, full Q above 20% output.
Fixed PF, full Q above 20% output.

Use capability table for sizing? Yes

Max reactive power, % of rated kVA (blank = N/A, operation not permitted)

V (pu) \ P loading (pu)	0	0.1	0.2	0.3
1.1	60	60	60	60
1.05	60	60	60	60
1	60	60	60	60
0.95	60	60	60	60
0.9	60	60	60	60
0.85	60	60	60	60

Sizing lookup (1.00 Vn row at PF loading):

Loading (= inverter PF)	0.918
q% at 1.00 Vn	35.75
q (kVAr) at 1.00 Vn	129.8

)

0.4	0.5	0.6	0.7	0.8	0.9	1
60	60	60	60	60	43.6	0
60	60	60	60	60	43.6	0
60	60	60	60	60	43.6	0
60	60	60	60	51.2	30.4	
60	60	60	56.6	41.2	0	
60	60	60	48.2	28.7		

COMPLIANCE		
Dispatch/invert	299.7	kW
Dispatch loadir	0.826	pu
Qmax required	17.37	MVAr

V-Q AT REGIS

V (pu)	q%/inv	Q prod (MVAı	Q abs (MVAr)	Req lag
1.10	55.8	30.62	43.11	0.00
1.05	55.8	30.62	43.11	0.00
1.00	55.8	30.62	43.11	17.37
0.95	45.9	24.07	36.56	17.37
0.90	30.7	14.01	26.50	0.00
0.85	0.0	-6.25	6.25	0.00

P-Q AT NOMI

P (pu)	P (MW)	disp (kW)	loading	q%/inv
0.00	0.0	0.0	0.000	60.0
0.05	2.6	15.0	0.041	60.0
0.10	5.3	30.0	0.083	60.0
0.15	7.9	45.0	0.124	60.0
0.20	10.6	59.9	0.165	60.0
0.25	13.2	74.9	0.206	60.0
0.30	15.8	89.9	0.248	60.0
0.35	18.5	104.9	0.289	60.0
0.40	21.1	119.9	0.330	60.0
0.45	23.8	134.9	0.372	60.0
0.50	26.4	149.8	0.413	60.0
0.55	29.0	164.8	0.454	60.0
0.60	31.7	179.8	0.495	60.0
0.65	34.3	194.8	0.537	60.0
0.70	37.0	209.8	0.578	60.0
0.75	39.6	224.8	0.619	60.0
0.80	42.2	239.7	0.660	60.0
0.85	44.9	254.7	0.702	60.0
0.90	47.5	269.7	0.743	60.0
0.95	50.2	284.7	0.784	60.0
1.00	52.8	299.7	0.826	55.8

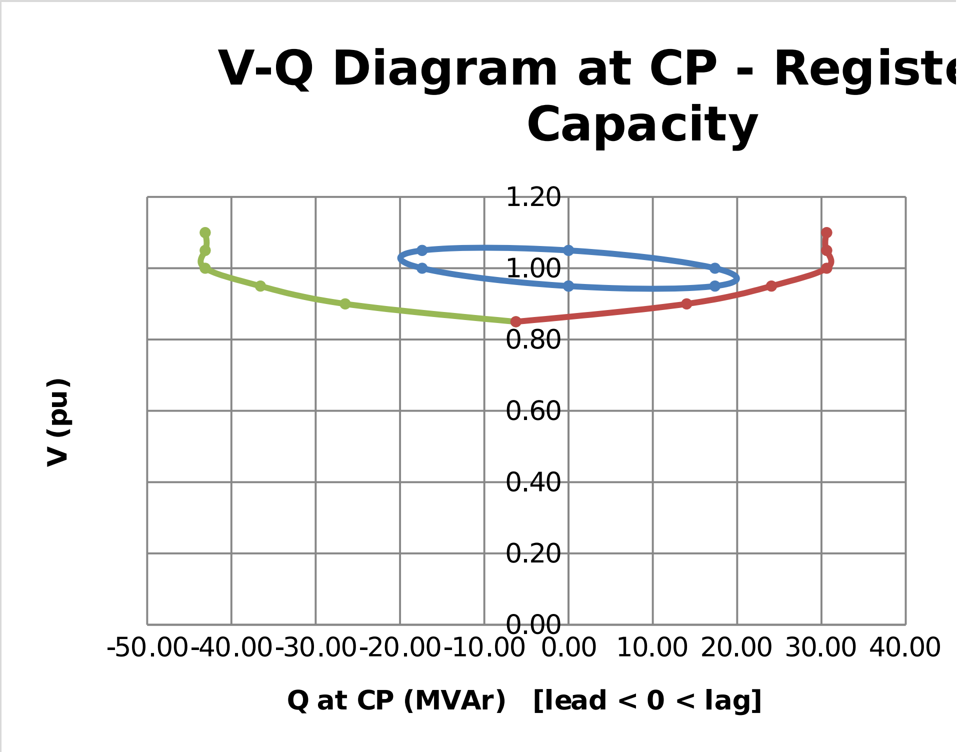
OVERALL VE **COMPLIANT**

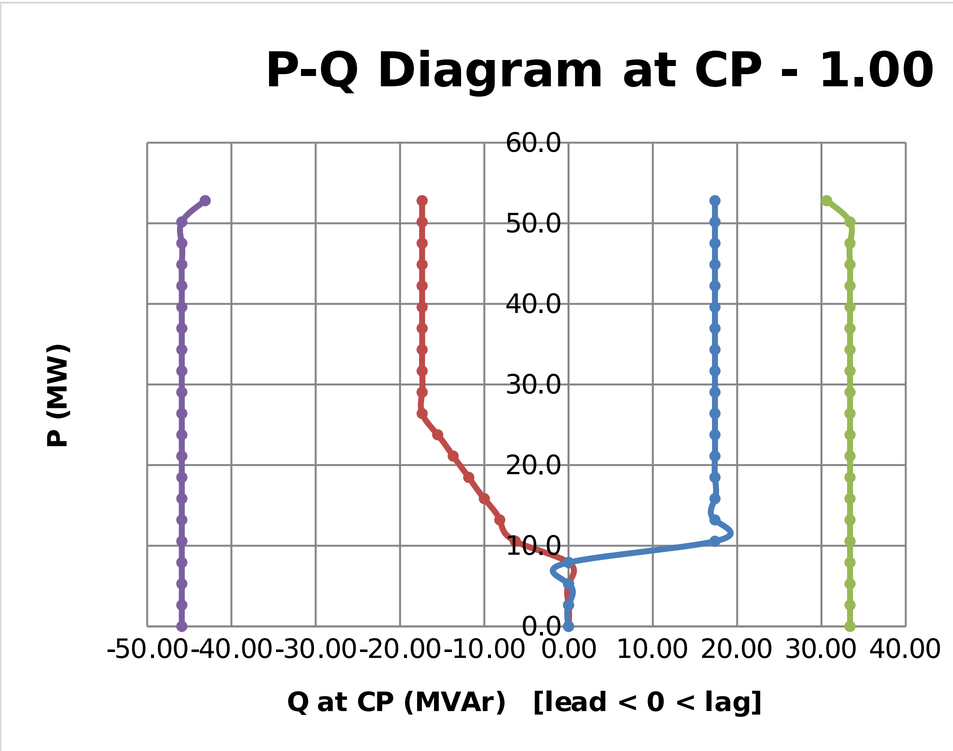
Req lead	Status
0.00	PASS
17.37	PASS
17.37	PASS
0.00	PASS
0.00	PASS
0.00	PASS

Req polygon

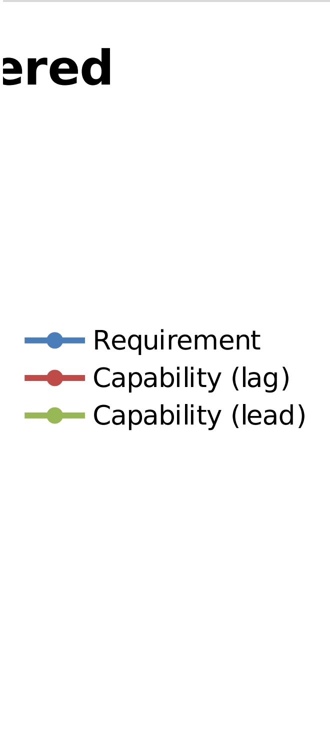
Q	V
0.00	0.95
17.37	0.95
17.37	1.00
0.00	1.05
-17.37	1.05
-17.37	1.00
0.00	0.95

Q prod	Q abs	Req lag	Req lead	Status
33.39	45.89	0.00	0.00	PASS
33.39	45.89	0.00	0.00	PASS
33.39	45.89	0.00	0.00	PASS
33.39	45.89	0.00	0.00	PASS
33.39	45.89	17.37	6.32	PASS
33.39	45.89	17.37	8.16	PASS
33.39	45.89	17.37	10.00	PASS
33.39	45.89	17.37	11.84	PASS
33.39	45.89	17.37	13.69	PASS
33.39	45.89	17.37	15.53	PASS
33.39	45.89	17.37	17.37	PASS
33.39	45.89	17.37	17.37	PASS
33.39	45.89	17.37	17.37	PASS
33.39	45.89	17.37	17.37	PASS
33.39	45.89	17.37	17.37	PASS
33.39	45.89	17.37	17.37	PASS
33.39	45.89	17.37	17.37	PASS
33.39	45.89	17.37	17.37	PASS
33.39	45.89	17.37	17.37	PASS
33.39	45.89	17.37	17.37	PASS
30.62	43.11	17.37	17.37	PASS





-Req lead	
0.00	
0.00	
0.00	
0.00	
-6.32	
-8.16	
-10.00	
-11.84	
-13.69	
-15.53	
-17.37	
-17.37	
-17.37	
-17.37	
-17.37	
-17.37	
-17.37	
-17.37	-45.89
-17.37	-45.89
-17.37	-45.89
-17.37	-43.11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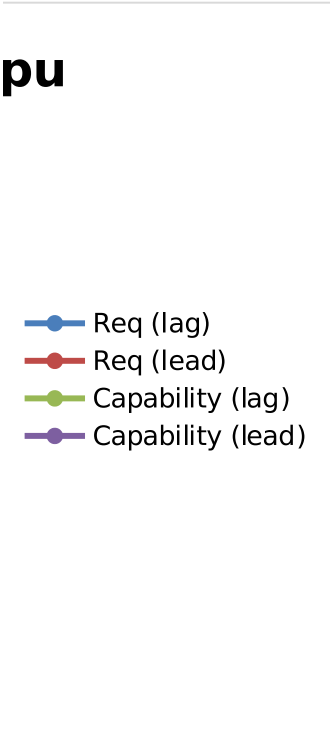
Requirement
Capability (lag)
Capability (lead)


pu
Req (lag)
Req (lead)
Capability (lag)
Capability (lead)